

ROBOTICS, DYNAMICS, AND CONTROLS

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.642
Control Systems for Mechanical
Engineering Applications

Fall

Advanced
Control Theory

535.724
Dynamics of Robots and
Spacecraft

Spring

535.630
Kinematics & Dynamics of Robots

Fall/Spring

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles

Fall

Advanced
Control Theory

ROBOTICS, DYNAMICS, AND CONTROLS

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 2 Courses

535.635
Introduction to Mechatronics

Spring

535.673
Mechanized Assembly: Hardware
and Algorithms

Summer

535.691
Haptic Interface Design

Fall

535.782
Haptic Applications

Spring/Summer

525.610
Microprocessors for Robotic
Systems

Spring

525.786
Human Robotics Interaction

Spring

Electronics

535.627
Computer-Aided Design

Every Term

535.612
Intermediate Dynamics

Fall

535.623
Intermediate Vibrations

Fall/Spring

535.630
Kinematics & Dynamics of Robots

Fall/Spring

535.724
Dynamics of Robots and
Spacecraft

Spring

535.628
Computer-Integrated Design &
Manufacturing

Spring

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles

Fall

525.661
UAV Systems and Control

Fall/Spring

605.613
Introduction to Robotics

Fall/Spring

605.716
Modeling and Simulation of
Complex Systems

Spring/Summer

Dynamics

535.638
Mechanical Packaging for
Electronics Systems

Summer/Fall

535.659
Manufacturing Systems Analysis

Fall

535.660
Precision Mechanical Design

Fall/Spring

535.603
Applied Optimal Control

Fall

535.622
Robot Motion Planning

Spring/Summer

535.741
Optimal Control and
Reinforcement Learning

Spring

665.681
Application of Sensing Systems

Every Term

535.642
Control Systems for Mechanical
Engineering Applications

Fall

535.645
Digital Control and Systems
Applications

Summer

525.609
Continuous Control Systems

Fall/Spring

525.626
Feedback Control in Biological
Signaling Pathways

Fall

525.645
Modern Navigation Systems

Fall/Spring

525.777
Control Systems

Fall

Intelligence & Planning

Mechanical Design

Control Theory